Project Overview

Hostnames, general configuration, what's built, and the full equipment list

AT A GLANCE

VLANS	CORE ↔ ACCESS UPLINK	WAN	HYPERVISOR	RACK POPULATED
5	**2×10G**	**DHCP**	**6 VMs**	**14U**
10 · 20 · 30 · 40 · 50	LACP Po1, Te1/0/1-2	From the STC modem, NAT on the ISR	Dell R720 · 384 GB RAM	of 22U · U5–U18

HOSTNAMES & ROLES

L2 Access Switch	**SWAGGER-ACCESS-SW01**
L3 Core Switch	**SWAGGER-CORE-SW01**
Firewall — ION 3000 (OPNsense)	**SWAGGER-FW01**
Edge Router — ISR 4331	**SWAGGER-4331-ISR01**
Hypervisor Host — Dell R720	**pve**
Proxmox FQDN	**pve.swagger.lab**
Storage — UniFi UNAS Pro	**SWAGGER-NAS01**

GENERAL CONFIGURATION

Domain Name	**swagger.lab**
IP Routing	**SWAGGER-CORE-SW01** Inter-VLAN routing at the core; OSPF Area 0 with OPNsense and the ISR
Firewall	**OPNsense on SWAGGER-FW01** Stateful filtering and Suricata IDS between the ISR and the core, no NAT
WAN	**DHCP on Gi0/0/1** STC HG8245Q modem upstream; the ISR NATs everything behind it
DNS / DHCP / NTP	**SWAGGER-DC-01** Relayed to VLAN 20 by the core; Guest and Cameras use core DHCP pools
Monitoring & SIEM	**SWAGGER-NMS-01** Wazuh, Splunk Free and LibreNMS containers
Automation	**Ansible** Ubuntu Linux control node, cisco.ios collection
Default Gateway Convention	**Last usable address** per subnet, e.g. 192.168.10.254

WHAT'S BUILT

Network

- Core and access switching, inter-VLAN routing, OSPF Area 0
- ISR 4331 edge routing and NAT overload
- 2×10G LACP uplink, core to access
- One UniFi AP with Internal and Guest SSIDs
- UniFi Protect cameras on VLAN 50

Servers & Storage

- Proxmox VE on the R720, 384 GB
- Two domain controllers: AD, DNS, DHCP, NTP
- File server with WSUS; CA with NPS
- Windows 11 domain client
- UniFi UNAS Pro for backups and footage

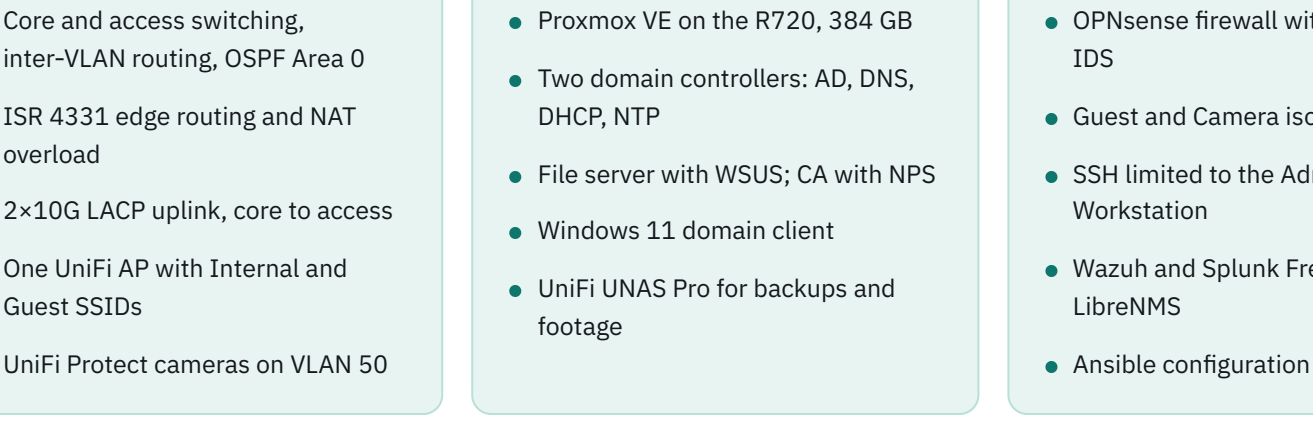
Security & Operations

- OPNsense firewall with Suricata IDS
- Guest and Camera isolation ACLs
- SSH limited to the Admin Workstation
- Wazuh and Splunk Free SIEM, LibreNMS
- Ansible configuration management

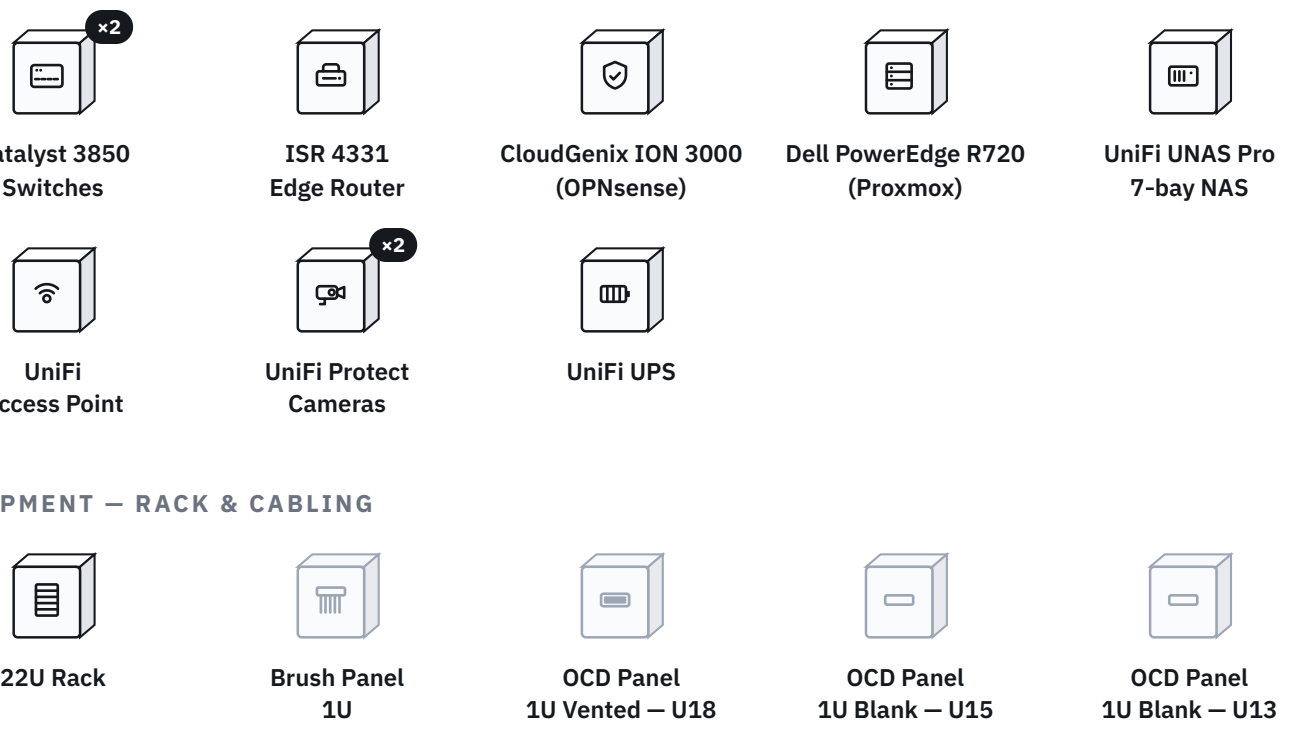
DOCUMENT MAP

01	**Project Overview**	This sheet
02	**Network Topology**	Every device, every link, every VLAN
03	**Logical Topology**	Layer-3 view: subnets, gateways, what routes to what
04	**VLAN & IP Addressing**	Subnets, DHCP scopes, static assignments
05	**Interfaces & Cabling**	Port-by-port mode, VLAN, and what's on the other end
06	**Rack Elevation**	Front & rear, every U, every cable run
07	**Security**	Trust zones, perimeter inspection, every control in place
08	**Services & Security Configuration**	The CLI: STP, OSPF, DHCP, ACLs, SSH, OPNsense, Ansible
09	**Proxmox Virtual Machines**	Host capacity and the full VM roster
10	**Docker Services**	Monitoring and SIEM containers on NMS-01

EQUIPMENT — ACTIVE



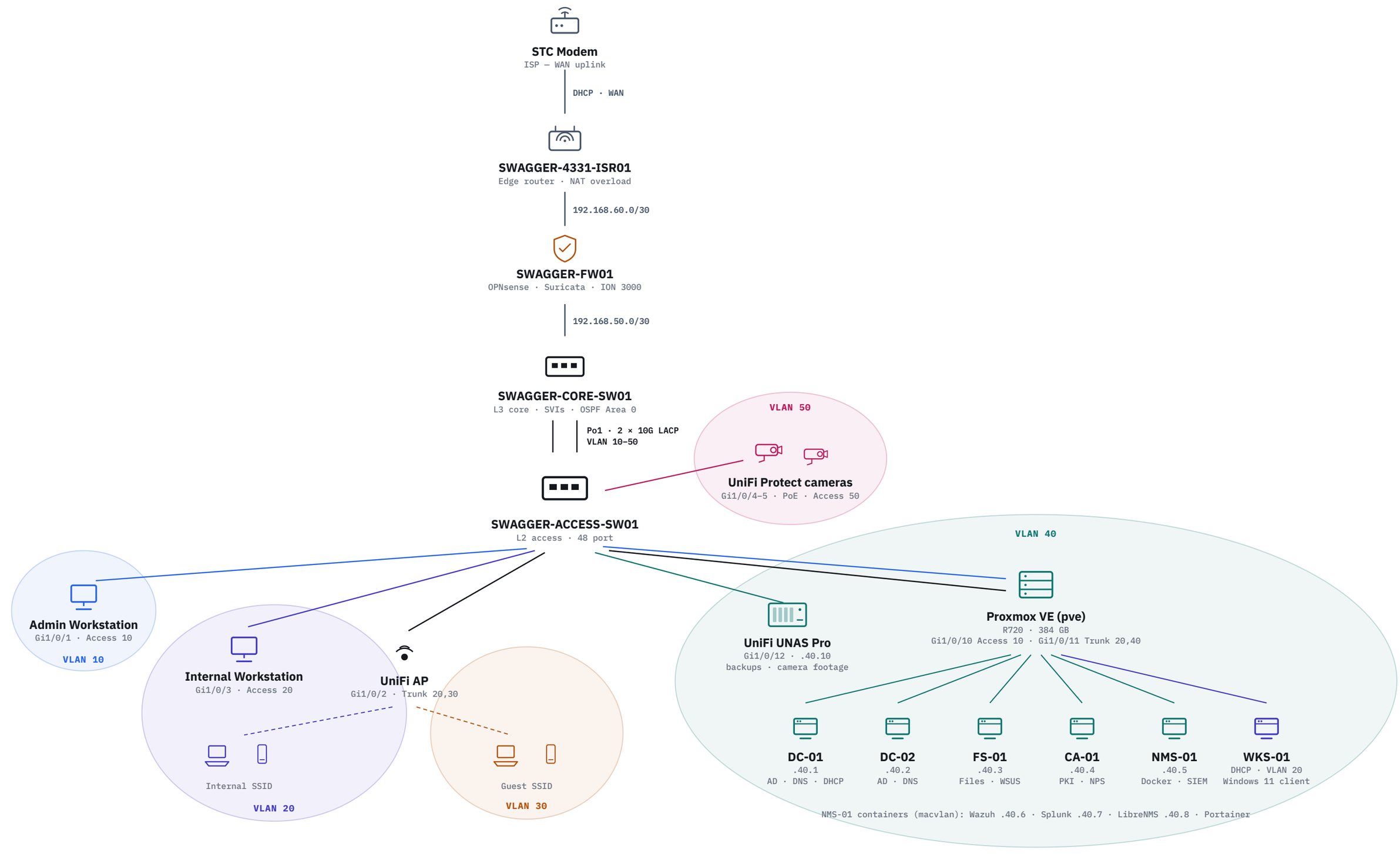
EQUIPMENT — RACK & CABLING



Network Topology

Every device, every link, and the VLAN each one lives on

VLAN 10 — Management · VLAN 20 — Internal · VLAN 30 — Guest · VLAN 40 — Servers · VLAN 50 — Cameras · Transit / WAN · Trunk · Wireless SSID



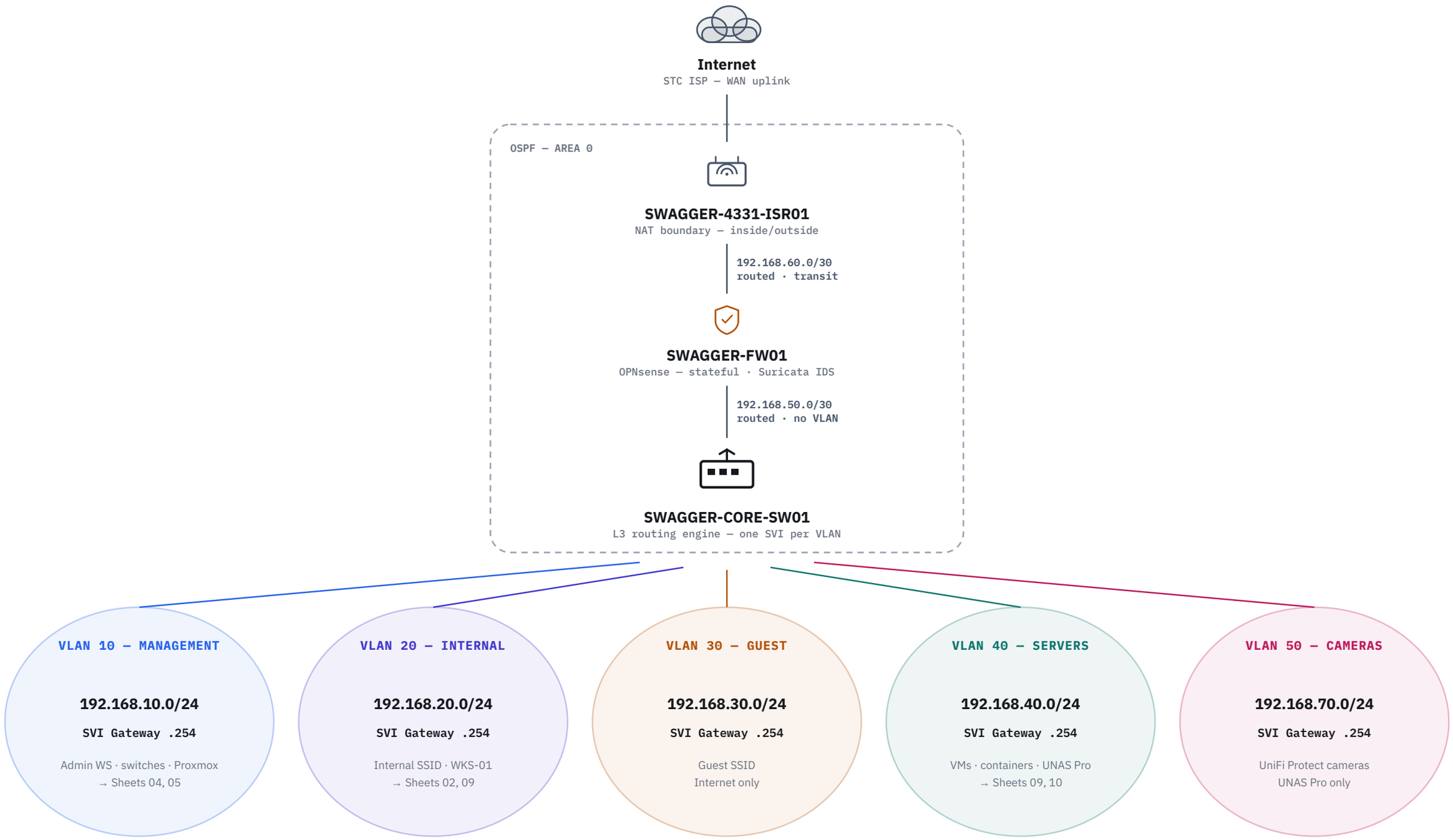
Ports match Sheet 05, addresses match Sheet 04, and the routed view of the same network is on Sheet 03.

One way in, one way out. OPNsense sits between two /30 transit links, so every packet entering or leaving the lab is filtered and inspected before the ISR translates it. Inside, the core routes between VLANs and the access switch only switches.

Logical Topology

Layer 3 view: subnets, gateways and routed paths, independent of physical cabling

> **This sheet is not the cabling diagram.** Sheet 02 (Network Topology) shows what plugs into what: ports, cables, every device. This sheet shows the other layer: how an IP packet gets from one subnet to another. SWAGGER-ACCESS-SW01 doesn't appear below because it's pure Layer 2; it forwards frames inside a VLAN and makes no routing decisions. Routing between VLANs happens in one place, the SVIs on SWAGGER-CORE-SW01, and anything leaving the network passes through OPNsense before it reaches the ISR.



Every line here is a routing relationship, not a cable. For the physical path, see Sheet 02.

INTER-VLAN ROUTING POLICY

FROM	TO	ROUTED?	NOTES
VLAN 10 — Mgmt	20, 40	Allowed	Admin reaches users and servers for management. Guest and Cameras stay sealed, even from here.
VLAN 20 — Internal	40	Allowed	AD, DNS and DHCP on SWAGGER-DC-01; file shares and WSUS on SWAGGER-FS-01.
VLAN 30 — Guest	10, 20, 40, 50	Blocked	ACL Guest-Isolation on SVI Vlan30. Internet only, with DHCP and DNS that never touch the domain.
VLAN 40 — Servers	10, 20	Allowed	Responses to clients, plus LibreNMS, Wazuh and Splunk reaching devices and agents.
VLAN 50 — Cameras	40	UNAS Pro only	ACL Cam-Isolation on SVI Vlan50. Footage goes to the UNAS Pro; no internet, no other VLANs.

> **Routing is open; leaving a sealed VLAN is not.** OSPF Area 0 advertises every subnet across the core, OPNsense and the ISR. ACLs on the Guest and Camera SVIs decide what may leave those two VLANs. Sheet 08 (Services) has the OSPF, DHCP and ACL config; Sheet 07 (Security) lists every control.

VLAN & IP Addressing

Subnet plan, assignment type, and gateways

VLAN & SUBNET PLAN

VLAN	NAME	SUBNET	ASSIGNMENT	GATEWAY
10	Management	192.168.10.0/24	Static	192.168.10.254
20	Internal	192.168.20.0/24	DHCP	192.168.20.254
30	Guest	192.168.30.0/24	DHCP	192.168.30.254
40	Servers	192.168.40.0/24	Static	192.168.40.254
50	Cameras	192.168.70.0/24	DHCP	192.168.70.254
999	Native	— (no hosts)	Unused	—
—	Transit · ISR ↔ FW	192.168.60.0/30	Static	— (transit)
—	Transit · FW ↔ Core	192.168.50.0/30	Static	— (transit)

HOST ADDRESSING

Management — VLAN 10

Admin Workstation	192.168.10.1
SWAGGER-ACCESS-SW01	192.168.10.2
Proxmox (vmbr0)	192.168.10.3
Default Gateway	192.168.10.254

Servers — VLAN 40

SWAGGER-DC-01	192.168.40.1
SWAGGER-DC-02	192.168.40.2
SWAGGER-FS-01	192.168.40.3
SWAGGER-CA-01	192.168.40.4
SWAGGER-NMS-01	192.168.40.5
Wazuh (container)	192.168.40.6
Splunk (container)	192.168.40.7
LibreNMS (container)	192.168.40.8
UniFi UNAS Pro	192.168.40.10
Default Gateway	192.168.40.254

Server VMs reach this VLAN through Proxmox vmbr1 (NIC2) → access switch Gi1/0/11. Containers join it through a macvlan network on SWAGGER-NMS-01. The UNAS Pro connects on Gi1/0/12.

Internal — VLAN 20

Assignment	DHCP relayed to SWAGGER-DC-01
DHCP Server	192.168.40.1
Default Gateway	192.168.20.254

Guest — VLAN 30

Assignment	DHCP pool on the core
DNS Server	1.1.1.1
Default Gateway	192.168.30.254

Cameras — VLAN 50

Assignment	DHCP pool on the core
Allowed Peer	UNAS Pro · 192.168.40.10
Default Gateway	192.168.70.254

Transit Links — /30

4331-ISR01 · Gi0/0/0	192.168.60.1
FW01 · WAN	192.168.60.2
FW01 · LAN	192.168.50.1
CORE-SW01 · Gi1/0/1	192.168.50.2

Interfaces & Cabling

Port-by-port mode, VLAN assignment, and what's on the other end

SWAGGER-ACCESS-SW01

L2 Access · 48-Port

PORT	MODE	VLAN	CONNECTS TO
Gi1/0/1	Access	10	Admin Workstation
Gi1/0/2	Trunk	20, 30	UniFi AP — Internal & Guest SSIDs, native 999
Gi1/0/3	Access	20	Internal workstation
Gi1/0/4–5	Access · PoE	50	UniFi Protect cameras
Gi1/0/6–9	Shutdown	—	Unused
Gi1/0/10	Access	10	Dell R720 NIC1 — Proxmox mgmt (vmbr0)
Gi1/0/11	Trunk	20, 40	Dell R720 NIC2 — Server VMs (vmbr1)
Gi1/0/12	Access	40	UniFi UNAS Pro — storage & camera footage
Gi1/0/13–48	Shutdown	—	Unused
Te1/0/1	Trunk · LACP	10,20,30,40,50	Po1 member 1 — SWAGGER-CORE-SW01
Te1/0/2	Trunk · LACP	10,20,30,40,50	Po1 member 2 — SWAGGER-CORE-SW01
Port-channel1	Trunk · LACP	10,20,30,40,50	**EtherChannel to SWAGGER-CORE-SW01**

SWAGGER-CORE-SW01

L3 Core · 48-Port

PORT	MODE	VLAN / ADDRESS	CONNECTS TO
Gi1/0/1	Routed	192.168.50.2/30	SWAGGER-FW01 LAN — firewall transit
Te1/0/1	Trunk · LACP	10,20,30,40,50	Po1 member 1 — SWAGGER-ACCESS-SW01
Te1/0/2	Trunk · LACP	10,20,30,40,50	Po1 member 2 — SWAGGER-ACCESS-SW01
Port-channel1	Trunk · LACP	10,20,30,40,50	**EtherChannel to SWAGGER-ACCESS-SW01**

SWAGGER-FW01

OPNsense · CloudGenix ION 3000

PORT	MODE	ADDRESS	CONNECTS TO
WAN	Routed	192.168.60.2/30	SWAGGER-4331-ISR01 Gi0/0/0 — edge transit
LAN	Routed	192.168.50.1/30	SWAGGER-CORE-SW01 Gi1/0/1 — core transit

SWAGGER-4331-ISR01

Edge Router · NAT

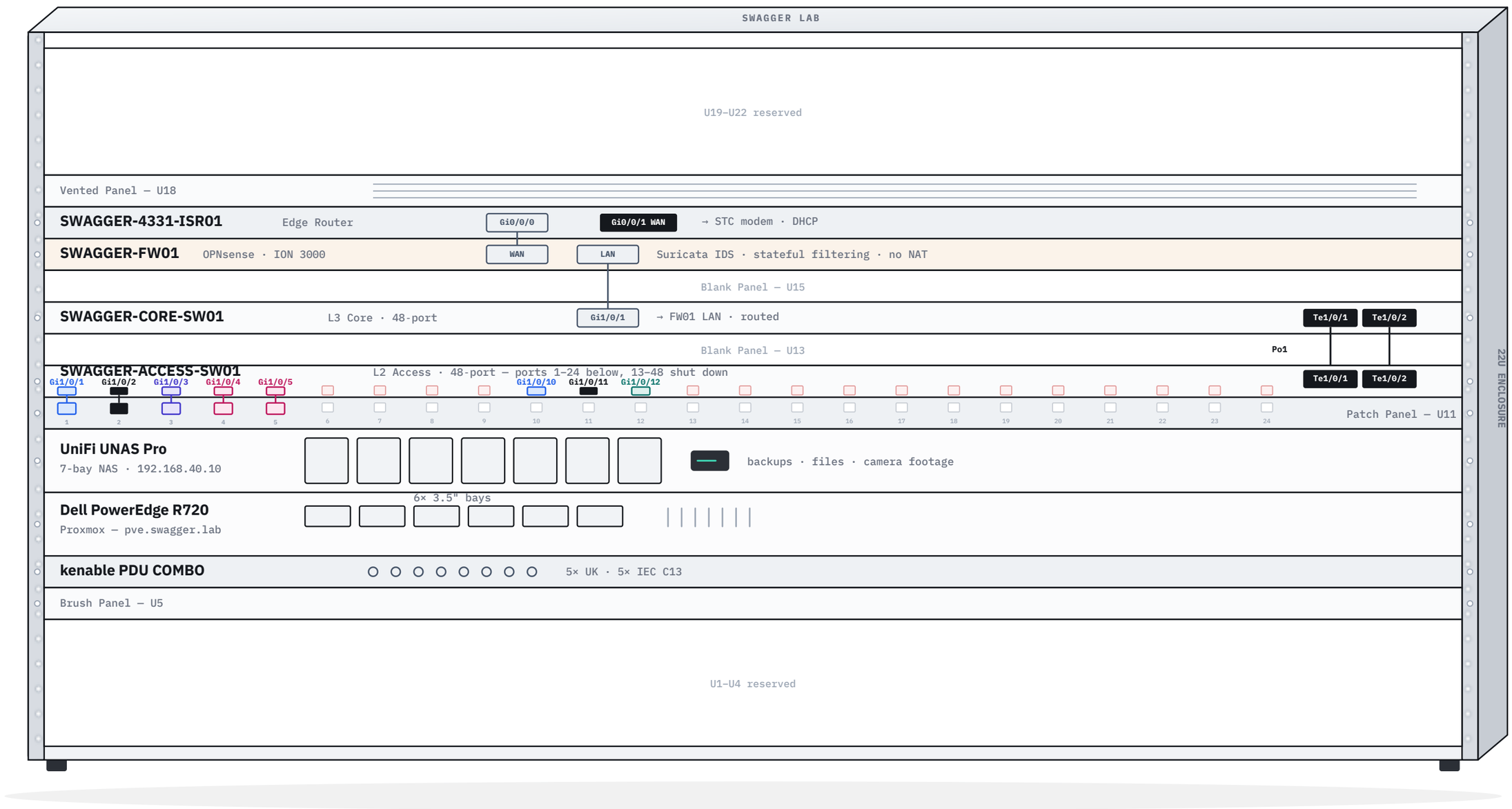
PORT	MODE	ADDRESS	CONNECTS TO
Gi0/0/0	Routed	192.168.60.1/30	SWAGGER-FW01 WAN — NAT inside
Gi0/0/1	DHCP · WAN	dynamic	STC modem — WAN uplink, NAT outside

Rack Elevation

22U enclosure, front & rear — to-scale, every port numbered and aligned to Sheet 05

VLAN 10 — Management · VLAN 20 — Internal · VLAN 40 — Servers · VLAN 50 — Cameras · Trunk · Shut down · Unused patch port · Power cord

FRONT

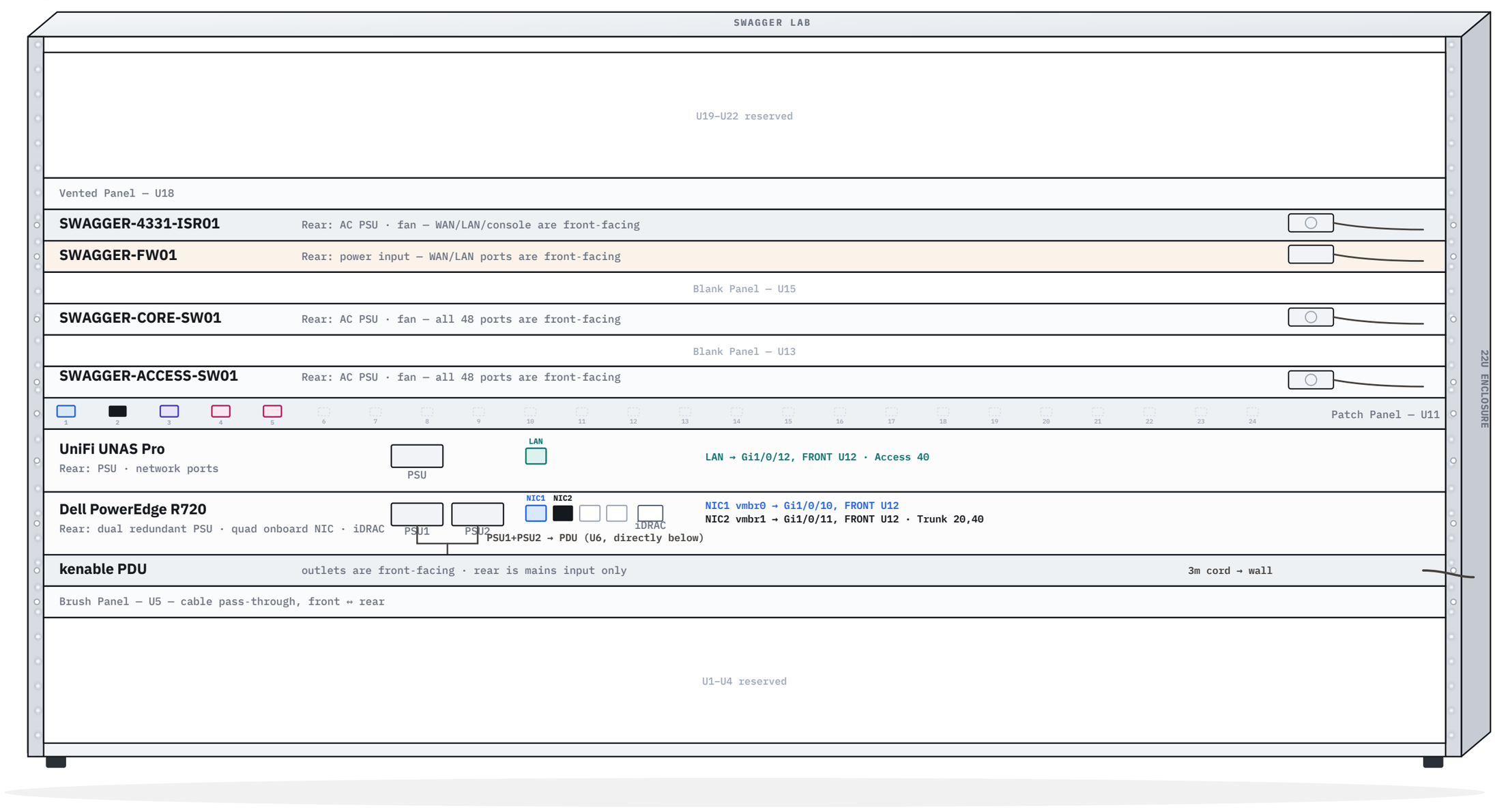


Patch ports 1–5 carry the in-wall runs and jump straight up to the matching switch port. Gi1/0/10–12 take direct cables from the R720 and the UNAS Pro instead. Live ports are colored by VLAN and named; full assignments are on Sheet 05.

ACCESS SWITCH — PORT ASSIGNMENT (MIRRORS SHEET 05)

PORT	MODE	VLAN	CONNECTS TO
Gi1/0/1	Access	10	Admin Workstation — in-wall run, patch port 1
Gi1/0/2	Trunk	20, 30	UniFi AP — Internal & Guest SSIDs, patch port 2
Gi1/0/3	Access	20	Internal workstation — patch port 3
Gi1/0/4–5	Access · PoE	50	UniFi Protect cameras — patch ports 4–5
Gi1/0/6–9	Shutdown	—	Unused
Gi1/0/10	Access	10	Dell R720 NIC1 — Proxmox mgmt (vmbr0) — rear, see below
Gi1/0/11	Trunk	20, 40	Dell R720 NIC2 — Server VMs (vmbr1) — rear, see below
Gi1/0/12	Access	40	UniFi UNAS Pro — rear, see below
Gi1/0/13–48	Shutdown	—	Unused
Te1/0/1, Te1/0/2	Trunk · LACP	10–50	**Po1 — SWAGGER-CORE-SW01 Te1/0/1, Te1/0/2 (2×0.5m 10G DAC)**

REAR



Patch ports 1–5 are punched down to wall jacks; the rest are capped with keystone blanks. The R720's two NICs and the UNAS Pro's LAN cable are the only rear-to-front runs in the rack.

REAR CONNECTOR → FRONT INTERFACE

REAR CONNECTOR	RESOLVES TO	DESTINATION
Patch panel — **Port 1**	**Gi1/0/1**	Admin Workstation — in-wall run
Patch panel — **Port 2**	**Gi1/0/2**	UniFi AP — in-wall run, PoE
Patch panel — **Port 3**	**Gi1/0/3**	Internal workstation — in-wall run
Patch panel — **Ports 4–5**	**Gi1/0/4–5**	UniFi Protect cameras — in-wall runs, PoE
Server — **NIC1 (vmbr0)**	**Gi1/0/10**	Proxmox management — ≈1m Cat6, routed via side channel
Server — **NIC2 (vmbr1)**	**Gi1/0/11**	Server VM trunk, VLAN 20 & 40 — ≈1m Cat6, routed via side channel
UNAS Pro — LAN	**Gi1/0/12**	**Storage and camera footage, VLAN 40 — ≈1m Cat6, routed via side channel**

Full path, rack to wall jack. The in-wall runs (Admin Workstation, UniFi AP, Internal workstation and both cameras) punch down at the rear on patch ports 1–5 and jump with short cords to Gi1/0/1–5, since the panel sits directly below the switch. The R720's NIC1 and NIC2 and the UNAS Pro's LAN port leave the rear and land on the *front* of SWAGGER-ACCESS-SW01 at Gi1/0/10–12. Every other port is shut down. Access and core aren't adjacent: a blank panel at U13 separates them, so the Te1/0/1-2 EtherChannel loops over it to form Po1. Every number here matches Sheet 05 — Interfaces & Cabling.

Security

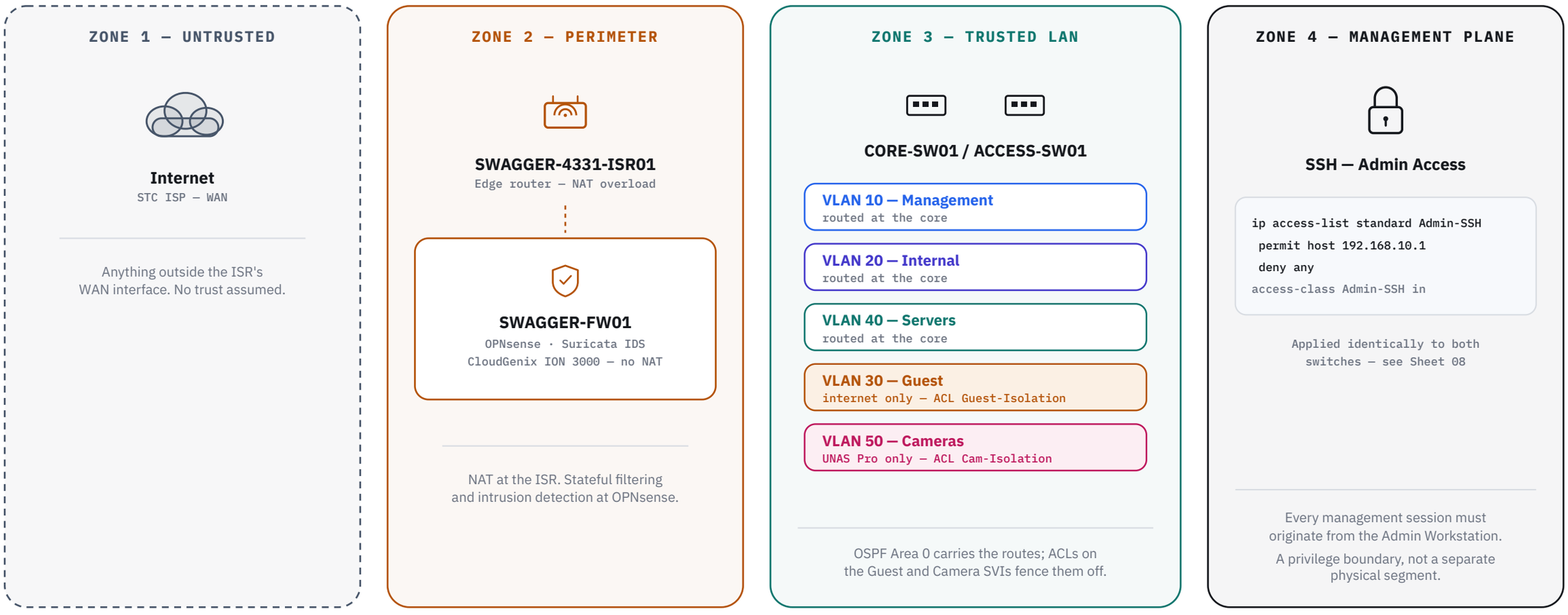
Trust zones, perimeter inspection, and the hardening in place

PERIMETER
OPNsense · Suricata
Stateful firewall and IDS behind the ISR

SEGMENTATION
5 VLANs · 2 ACLs
Guest and Cameras fenced off at the core

MANAGEMENT ACCESS
SSH · ACL-restricted
Admin Workstation only, both switches

MONITORING
Wazuh · Splunk · LibreNMS
Containers on SWAGGER-NMS-01 — Sheet 10



Four zones, left to right, least trusted to most trusted. Zone 4 overlaps Zone 3 physically: it's a privilege boundary enforced by ACL, not a separate cable path.

HARDENING CHECKLIST

CONTROL	STATUS	DETAIL	SHEET
WAN NAT boundary	Live	NAT overload on SWAGGER-4331-ISR01, outside on Gi0/0/1. OPNsense adds no NAT of its own.	05
Perimeter firewall	Live	OPNsense on SWAGGER-FW01 between the ISR and the core. Stateful, default deny inbound.	02, 07
Intrusion detection	Live	Suricata inspects all traffic crossing SWAGGER-FW01.	07
SSH management ACL	Live	vty 0–15 restricted to host 192.168.10.1 (Admin WS) on both switches.	08
Unused port shutdown	Live	Gi1/0/6–9 and Gi1/0/13–48 administratively down on SWAGGER-ACCESS-SW01.	05, 06
VTP / trunk hardening	Live	VTP transparent mode; switchport nonegotiate on Po1; native VLAN 999.	08
STP root protection	Live	SWAGGER-CORE-SW01 is root primary for VLANs 10/20/30/40/50.	08
VLAN segmentation	Live	5 broadcast domains (10/20/30/40/50), routed at the core.	03, 04
Guest isolation	Live	ACL on SVI Vlan30: internet only, no private destinations.	03, 08
Camera isolation	Live	ACL on SVI Vlan50: cameras reach the UNAS Pro and nothing else.	03, 08
SIEM / log monitoring	Live	Wazuh and Splunk Free containers on SWAGGER-NMS-01.	10
SNMP network monitoring	Live	LibreNMS polls every switch, router and the firewall.	10

A homelab, built to production habits. Least-privilege management access, segmentation, inspection at the perimeter and central logging. Not full zero-trust, but every layer has a control and every control is listed above.

Services & Security

Spanning tree, routing, DHCP, firewall, hardening & automation

GLOBAL — BOTH SWITCHES

```
vtp mode transparent
switchport nonegotiate  ·  on Po1, SWAGGER-ACCESS-SW01
```

SWAGGER-CORE-SW01

Spanning Tree

```
spanning-tree vlan 10 root primary
spanning-tree vlan 20 root primary
spanning-tree vlan 30 root primary
spanning-tree vlan 40 root primary
spanning-tree vlan 50 root primary
```

OSPF — Process 1, Area 0

```
network 192.168.10.254 0.0.0.255 area 0
network 192.168.20.254 0.0.0.255 area 0
network 192.168.30.254 0.0.0.255 area 0
network 192.168.40.254 0.0.0.255 area 0
network 192.168.70.254 0.0.0.255 area 0
network 192.168.50.2 0.0.0.3 area 0
PASSIVE-INTERFACE
VLAN 10, 20, 30, 40, 50
```

DHCP

```
RELAY — DOMAIN VLAN
VLAN 20 → ip helper-address 192.168.40.1 · DC-01
LOCAL POOLS — ISOLATED VLANS
ip dhcp pool GUEST
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.254
 dns-server 1.1.1.1
ip dhcp pool CAMERAS
 network 192.168.70.0 255.255.255.0
 default-router 192.168.70.254
```

Inter-VLAN ACLs

```
ip access-list extended Guest-Isolation
 permit udp any any eq bootps
 deny ip any 192.168.0.0 0.0.255.255
 permit ip any any
ip access-list extended Cam-Isolation
 permit udp any any eq bootps
 permit ip any host 192.168.40.10
 deny ip any any
Vlan30 → ip access-group Guest-Isolation in
Vlan50 → ip access-group Cam-Isolation in
```

SWAGGER-4331-ISR01

Static Route

```
ip route 0.0.0.0 0.0.0.0 192.168.100.1
```

OSPF — Process 1, Area 0

```
network 192.168.60.1 0.0.0.0 area 0
default-information originate
```

WAN Addressing

```
interface Gi0/0/1 → ip address dhcp
description TO-STC-MODEM-WAN-DHCP
```

NAT Overload

```
interface Gi0/0/0 → ip nat inside
interface Gi0/0/1 → ip nat outside
ACL — STANDARD NAT
permit 192.168.10.0 0.0.0.255
permit 192.168.20.0 0.0.0.255
permit 192.168.30.0 0.0.0.255
permit 192.168.40.0 0.0.0.255
permit 192.168.60.0 0.0.0.3 (firewall updates)
ip nat inside source list NAT interface Gi0/0/1
overload
```

SWAGGER-FW01 · OPNSENSE

Interfaces

```
WAN → 192.168.60.2/30 (to ISR Gi0/0/0)
LAN → 192.168.50.1/30 (to core Gi1/0/1)
```

Routing — FRR, OSPF Area 0

```
network 192.168.60.0/30 area 0
network 192.168.50.0/30 area 0
OUTBOUND NAT
Disabled · NAT stays on the ISR
```

Rules & Inspection

```
WAN → default deny inbound, stateful
LAN → allow internal networks out
IDS → Suricata on WAN and LAN
GUI → reachable from 192.168.10.1 only
```

SSH — CORE, ACCESS & ISR

```
GLOBAL CONFIGURATION
username <ADMIN_USER> privilege 15 secret
<SECURE_PASSWORD>
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
```

```
VTY LINES & ACCESS CONTROL
line vty 0 15
 transport input ssh
 login local
 access-class Admin-SSH in
ip access-list standard Admin-SSH
 permit host 192.168.10.1
 deny any
```

AUTOMATION — ANSIBLE

```
CONTROL NODE
Ubuntu Linux · ansible-core
collection cisco.ios · network_cli over SSH
Held to the Admin-SSH ACL like any session
```

```
INVENTORY
[switches] CORE-SW01, ACCESS-SW01
[routers]  4331-ISR01
PLAYBOOKS
config backup · VLAN & trunk baseline
SSH & NTP baseline
```

Proxmox Virtual Machines

Hypervisor host and the virtual machine roster

HYPERVISOR HOST



pve.swagger.lab

Dell PowerEdge R720 — Proxmox VE hypervisor

VMBR0 — MGMT	VMBR1 — TRUNK 20,40
192.168.10.3	Gi1/0/11

MEMORY	PROCESSORS	GUESTS	BACKUPS
384 GB	**Dual CPUs**	**6 VMs**	**UNAS Pro**
Every VM runs at once	Intel Xeon E5, two sockets	5 servers, 1 client	Off-host, over VLAN 40

VIRTUAL MACHINE ROSTER



SWAGGER-DC-01

LIVE

Primary Domain Controller

AD DS · DNS · DHCP · NTP

Windows Server 2022
192.168.40.1 — VLAN 40



SWAGGER-NMS-01

LIVE

Docker Host, Monitoring & SIEM

Docker CE · Portainer · LibreNMS · Wazuh · Splunk

Linux
192.168.40.5 — VLAN 40



SWAGGER-DC-02

LIVE

Secondary Domain Controller

AD DS · DNS

Windows Server 2022
192.168.40.2 — VLAN 40



SWAGGER-FS-01

LIVE

File Server & Patching

File Server · WSUS

Windows Server 2022
192.168.40.3 — VLAN 40



SWAGGER-CA-01

LIVE

PKI & Network Policy

Certificate Authority · NPS

Windows Server 2022
192.168.40.4 — VLAN 40



SWAGGER-WKS-01

LIVE

Domain-Joined Test Client

GPO Testing

Windows 11
DHCP — VLAN 20

WKS-01 is a VM, not a physical PC. It runs on the same Proxmox host as the servers, but its vNIC sits on vmbr1's VLAN 20 leg, so logically it behaves as just another Internal client, addressed by DHCP from SWAGGER-DC-01.

Docker Services

Containerized monitoring and SIEM stack on SWAGGER-NMS-01

SWAGGER-NMS-01
Docker host VM on Proxmox · full record on Sheet 09

192.168.40.5 — VLAN 40 LIVE

CONTAINER NETWORK

DRIVER	SUBNET	GATEWAY
macvlan	**192.168.40.0/24**	**192.168.40.254**
vlan40-net, parent eth0	Same segment as the servers	SVI on SWAGGER-CORE-SW01

CONTAINER STACK

Wazuh

192.168.40.6 LIVE

SIEM: log analysis, file integrity monitoring, intrusion detection

`wazuh-manager + indexer + dashboard` `:443 dashboard` `:1514/1515 agents` `:514 syslog`

Agents run on every Windows Server VM. OPNsense, the ISR and both switches send syslog, so Suricata alerts and device events land in one place.

Splunk Free

192.168.40.7 LIVE

Search, dashboards and reports over the same log feeds

`splunk/splunk` `:8000 web` `:9997 forwarders` `:514 syslog`

A second lens on the data: Splunk's search language for hunting and dashboards. The free license's 500 MB a day covers the lab's volume.

LibreNMS

192.168.40.8 LIVE

SNMP network monitoring: device health, interface counters, uptime

`librenms/librenms` `:8000 web` `SNMP v2c/v3 poll`

Polls both Catalyst 3850s, the ISR 4331 and OPNsense, with alerting when a link or device goes down.

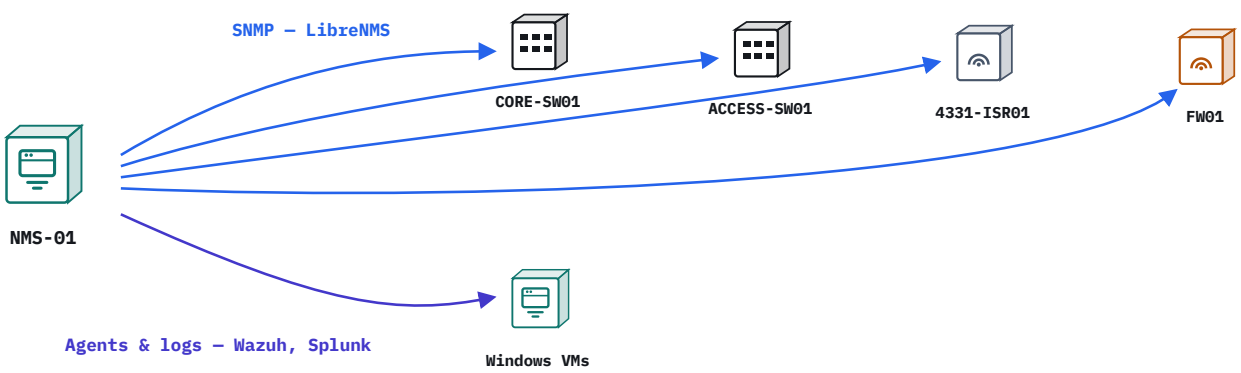
Portainer

192.168.40.5 LIVE

Web UI for managing every container on the host

`portainer/portainer-ce` `:9443 https` `host network`

MONITORING REACH



One IP per service. The macvlan network puts Wazuh, Splunk and LibreNMS straight onto VLAN 40 with their own addresses, so firewall rules, ACLs and logs see each service on its own instead of one shared host.